Project Title
(SmartSpell)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Describe what your project presents within 150 words.

This project focuses on developing an intelligent spell checker that helps users find the correct spelling of misspelled words. The system takes a dictionary and a word entered by the user as input. It compares the misspelled word with the words in the dictionary using Levenshtein Edit Distance, which calculates the minimum number of changes needed to convert one word into another. Based on this distance, the system ranks the dictionary words and provides the most similar words as suggestions. The project is designed to work with large dictionaries while keeping the response time low. It also supports Unicode text and multiple languages, making it useful for different types of users and applications. The main concept behind the project is Dynamic Programming, which is used to calculate edit distances efficiently. Overall, the project demonstrates how a simple algorithm can be applied to build a practical and useful spell-checking system for real-world text processing.

Signature of the Guide